

of Questions: 30

Question #: 1

During childbirth, an obstetrician may elect to perform a pudendal nerve block to anesthetize the perineum and vagina. This can be done transvaginally by injecting the anesthetic near the pudendal nerve as it passes from the pelvis into the perineum. To locate the nerve, the physician inserts a finger into the vagina and palpates laterally to most likely locate which of the following landmarks?

- A. Ischial tuberosity
 - B. Anterior superior iliac spine
 - C. Ischial spine
 - D. Posterior fornix
 - E. Posterior superior iliac spine
-

Question #: 2

A 27-year-old woman is brought to the emergency department by the paramedics because of 1-hour history of motor vehicle collision. Physical examination shows tender urinary bladder, loss of anal tone and perianal numbness. A urinary catheter is inserted to relieve urinary retention. Imaging studies show multiple fractures of pelvis and lower lumbar vertebrae. Which of the following nerves was most likely involved to produce urinary symptoms?

- A. Superior hypogastric
- B. Pelvic splanchnic
- C. Pelvic splanchnic
- D. Lumbar splanchnic

E. Pudendal

Question #: 3

A 1-week-old boy is brought to the physician by his mother for postnatal examination. Physical examination shows bilaterally testes are located in the inguinal canal. No other abnormalities noted. Which of the following structures is most likely facilitates the descent of the testes to the scrotal sac?

- A. Peritoneal evagination into the scrotal sac
 - B. Remnant of the processus vaginalis
 - C. Fascial extensions of the inguinal abdominal wall
 - D. Tissue suspending the testes from the mesonephros
 - E. Cord extending between the testis and scrotum
-

Question #: 4

A 30-year-old woman comes to the physician for a prenatal checkup. The patient has not used cigarettes or alcohol and plans to continue abstaining during pregnancy. Vital signs are within normal limits. Ultrasonography shows a deficient amount of amniotic fluid. Which of the following most likely contributed to this condition?

- A. Maternal diabetes
 - B. Fetal microencephaly
 - C. Fetal anencephaly
 - D. Fetal renal agenesis
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Question #: 5

A 35-year-old woman G2 P1 is 30 weeks pregnant, comes to the

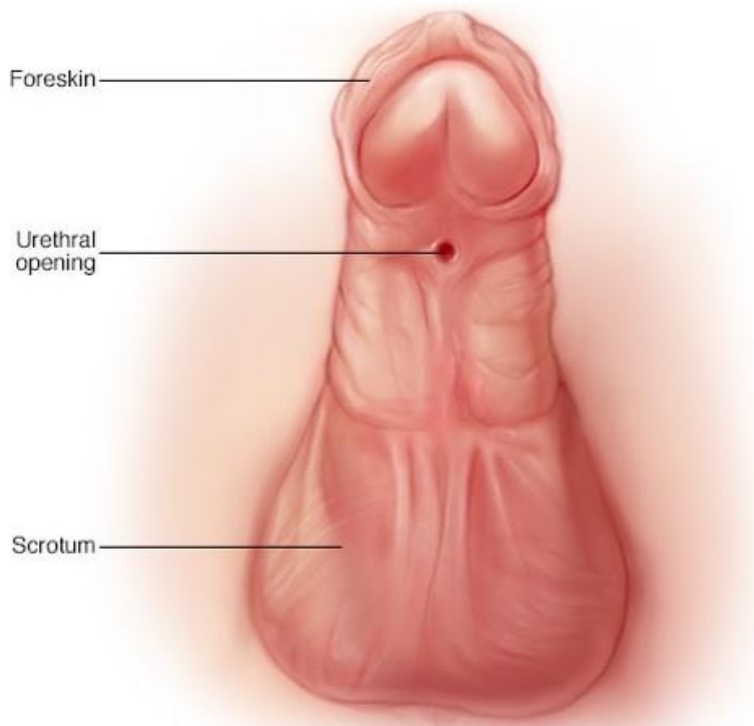
emergency department because of metrorrhagia (non-menstrual vaginal bleeding) for past 4 hours. Physical examination shows the patient to be normotensive with discolored conjunctiva. Doppler ultrasonographic examination shows a non-homogenous placenta protruding into the bladder. Which of the following most correctly describes the patient's condition?

- A. Placenta accreta
 - B. Placenta increta
 - C. Placenta percreta
 - D. Placenta previa
-

Question #: 6

During a newborn checkup on a baby boy, a physician discovers the external urethral orifice is located in an ectopic location as shown in the attached diagram. Which of the following is the most likely diagnosis?

- A. Epispadias
- B. Cryptorchidism
- C. Perineal hypospadias
- D. Penile hypospadias
- E. Hydrocele



Question #: 7

A 36-year-old man comes to the emergency department because of a 12-hour history of perineal pain. He has difficulty urinating and says there is blood in his semen during ejaculation. Imaging studies show enlarged and inflamed seminal vesicles. Which of the following nerves transmits the visceral pain fibers from this organ back to the spinal cord?

- A. Lumbar splanchnics
- B. Pelvic splanchnics
- C. Pudendal
- D. Ilioinguinal
- E. Obturator

Question #: 8

A 20-year-old woman comes to the physician because of painful, heavy bleeding during her last menstrual cycle. An ultrasound examination shows endometriomas (ectopic endometrial tissue) of the cervical canal and the uterus in the normal anteverted position. The angle between which of the following organs best describes this position?

- A. Vagina and the cervix
 - B. Uterine body and cervix
 - C. Uterine body and bladder
 - D. Cervix and the rectum
 - E. Vagina and uterine body
-

Question #: 9

A 54-year-old man comes to the physician for a follow-up examination. He has a 2-year history of type 2 diabetes mellitus managed with metformin. He feels well and reports no current symptoms. He does not use tobacco, alcohol, or recreational drugs. Blood pressure is 120/74 mm Hg, pulse is 62/min, and BMI is 28 kg/m². Physical examination is unremarkable. Laboratory studies show hemoglobin A1c is 8.6% (Reference range: <6%). Because glycemic control remains suboptimal, the patient is prescribed pioglitazone, a hypoglycemic agent that enhances insulin sensitivity. Which of the following mechanisms best explains the desired effect of this medication?

- A. Decreased expression of GLUT-4
- B. Decreased expression of glucagon like peptide-1 (GLP-1)
- C. Increased hepatic glucose production
- D. Increased production of adiponectin
- E. Increased production of interleukin-6

Question #: 10

A 38-year-old woman comes to the physician because of a 2-month history of progressive darkening of her skin. She reports generalized fatigue and unintentional weight loss. Blood pressure is 100/70 mm Hg. Physical examination shows diffuse hyperpigmentation of the skin, including the palmar creases and buccal mucosa. Laboratory studies show elevated plasma ACTH concentration and the presence of adrenocortical antibodies. Which of the following findings is most likely present in this patient?

- A. Increased renal sodium reabsorption
- B. Elevated serum sodium levels
- C. Increased renal calcium reabsorption
- D. Decreased renal potassium excretion
- E. Elevated plasma glucose levels

Question #: 11

A 2-week-old infant is brought to the emergency department because of a 6-hour history of poor feeding and lethargy. The parents report that the infant has had vomiting and decreased urine output over the past day. Blood pressure is 70/40 mm Hg. The infant appears severely dehydrated. Physical examination shows ambiguous genitalia. Laboratory studies are shown in the table. Which of the following enzyme deficiencies is most likely responsible for this patient's condition?

- A. 3-beta hydroxysteroid dehydrogenase
- B. 17-hydroxylase (CYP 17)
- C. 21-hydroxylase (CYP 21)
- D. 11-hydroxylase (CYP 11B)

E. Aromatase (CYP 19)

Laboratory Test	Result	Reference Range
Serum sodium	126 mEq/L	136-146 mEq/L
Serum potassium	5.8 mEq/L	3.5-5.0 mEq/L
Serum glucose	45 mg/dL	70-100 mg/dL
Plasma renin activity	Increased	—
Serum cortisol	Decreased	—
Serum 17-hydroxyprogesterone	Increased	—

Question #: 12

A 50-year-old man comes to the physician because of a 2-month history of intermittent episodes of lightheadedness and fainting. The episodes typically occur in the morning and improve after drinking a glass of orange juice. BMI is 21 kg/m², and the patient has lost 4.5 kg (10 lb) since the previous visit 4 months ago. Laboratory studies show fasting serum glucose concentration is 60 mg/dL (reference range: 70–100 mg/dL). Abdominal imaging shows a mass involving specific cells of the pancreatic islets of Langerhans. Additional laboratory studies show elevated concentrations of C-peptide and a specific peptide hormone. Which of the following best describes the signal transduction mechanism of this hormone?

- A. Activates protein kinase A
- B. Increases intracellular calcium levels
- C. Activates cGMP-dependent protein kinase
- D. Binds to intracellular receptors
- E. Activates receptor tyrosine kinase

Question #: 13

A 45-year-old man comes to the physician because of a 6-month history of unintentional weight loss despite an increased appetite. He has lost 4.5 kg (10 lb) since his previous visit 6 months ago. His BMI is 22 kg/m². Laboratory studies show fasting serum glucose concentration is 200 mg/dL (reference range: 70-100 mg/dL). Abdominal imaging shows a mass involving specific cells of the pancreatic islets of Langerhans. Activation of which of the following intracellular signaling proteins is most likely mediated by the hormone secreted by this tumor?

- A. Adenylate cyclase
 - B. Protein kinase C
 - C. cGMP-dependent protein kinase
 - D. Phosphodiesterase
 - E. Tyrosine kinase
 - F. Guanylate cyclase
-

Question #: 14

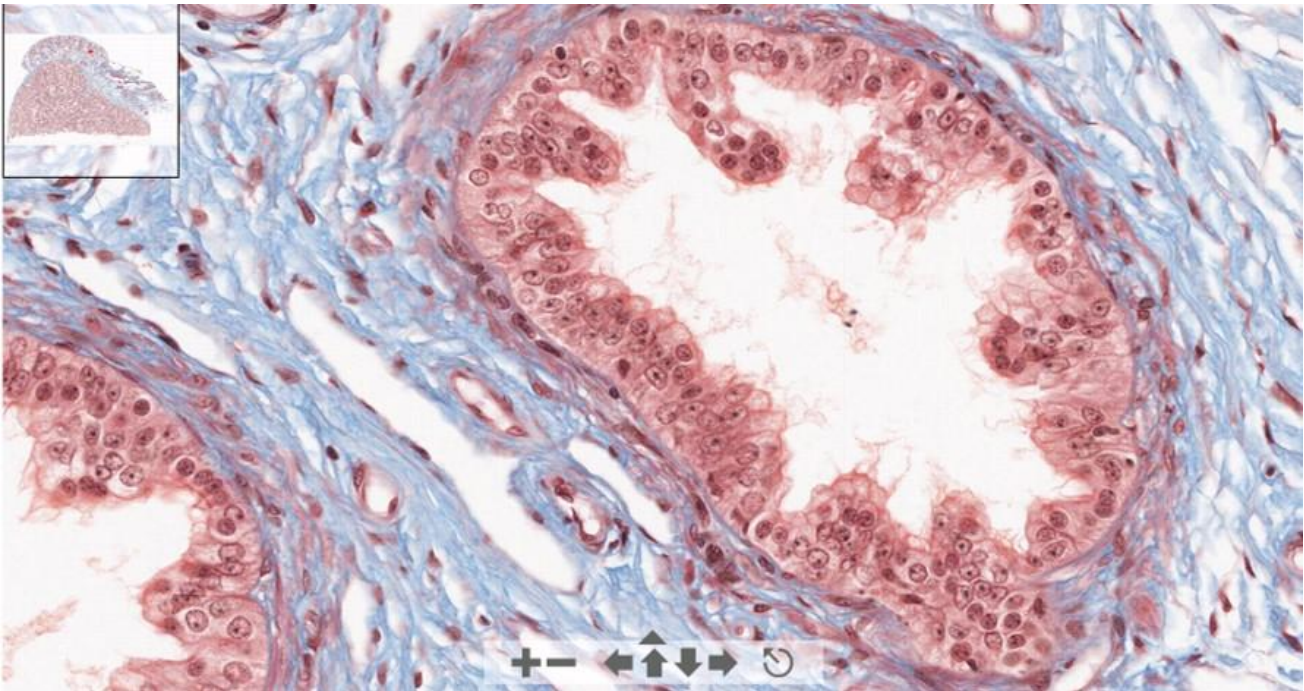
A 12-year-old girl is brought to the physician by her mother because of a 1-month history of weight loss, fatigue, excessive thirst, and urination. Her mother also reports burning during urination. There is no family history of major medical illnesses. Physical examination shows lethargy, dry oral mucosa, and decreased skin turgor. Which of the following laboratory findings is most likely in this patient?

- A. Plasma glucose of 100 mg/dl
 - B. Urine glucose absent
 - C. Plasma glucose of 250 mg/dl
 - D. Plasma glucose of 40 mg/dl
 - E. HbA1c of 5%
-

Question #: 15

A 32-year-old man comes to the physician because of 2-week history of pain and swelling in his left scrotum. A biopsy specimen from the left scrotal swelling is shown in the attached photomicrograph. Which of the following functions is most likely affected due to damage of this structure?

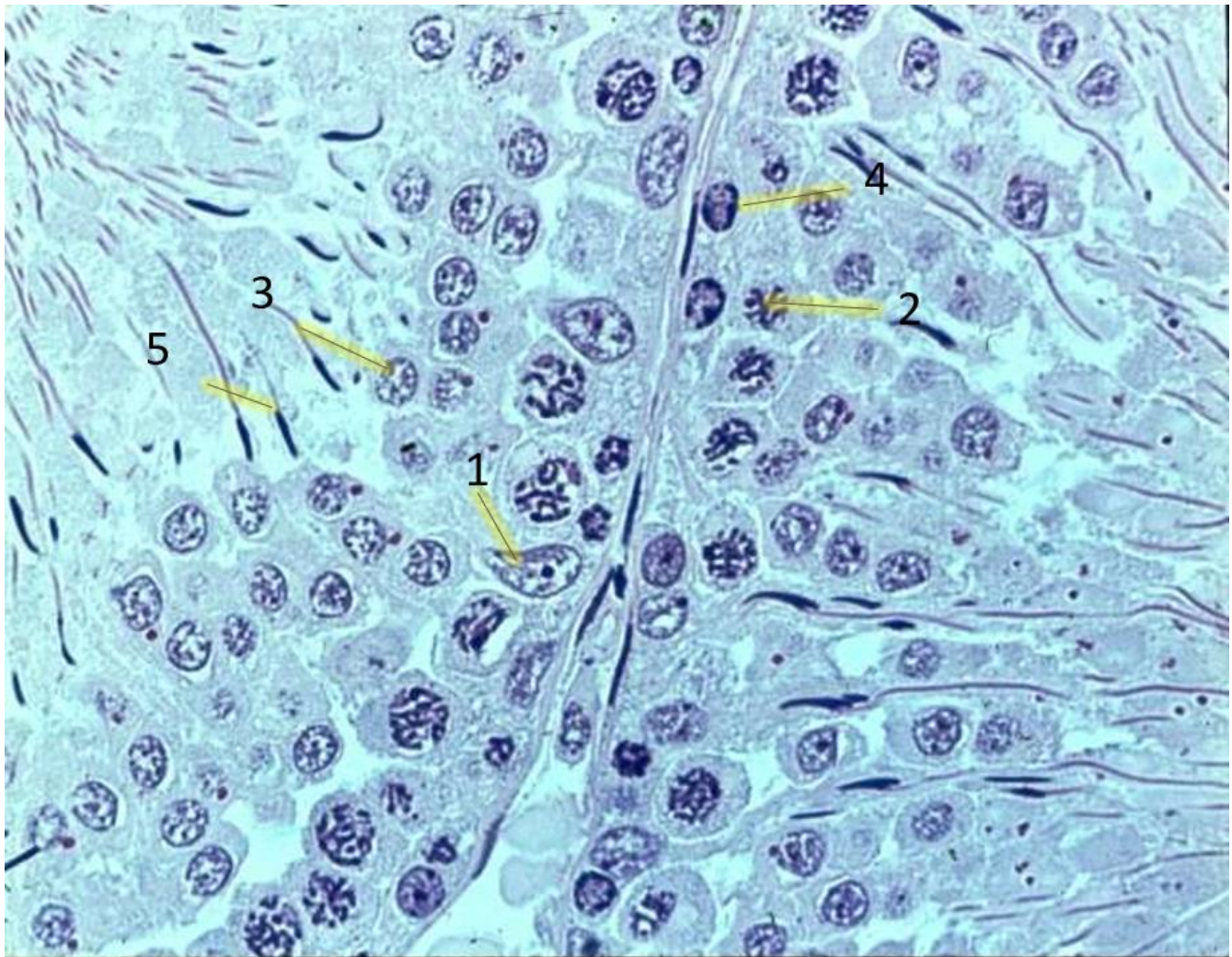
- A. Blood-testis barrier
- B. Reabsorption of seminiferous fluid
- C. Testosterone production
- D. Sperm storage
- E. Reduced semen volume production



Question #: 16

A researcher is investigating the effects of a novel compound on cells of the seminiferous epithelium. He discovers that the compound affects spermatogenesis, particularly cells that are in their first phase of meiotic division. Which of the following cells are most likely affected?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5



Question #: 17

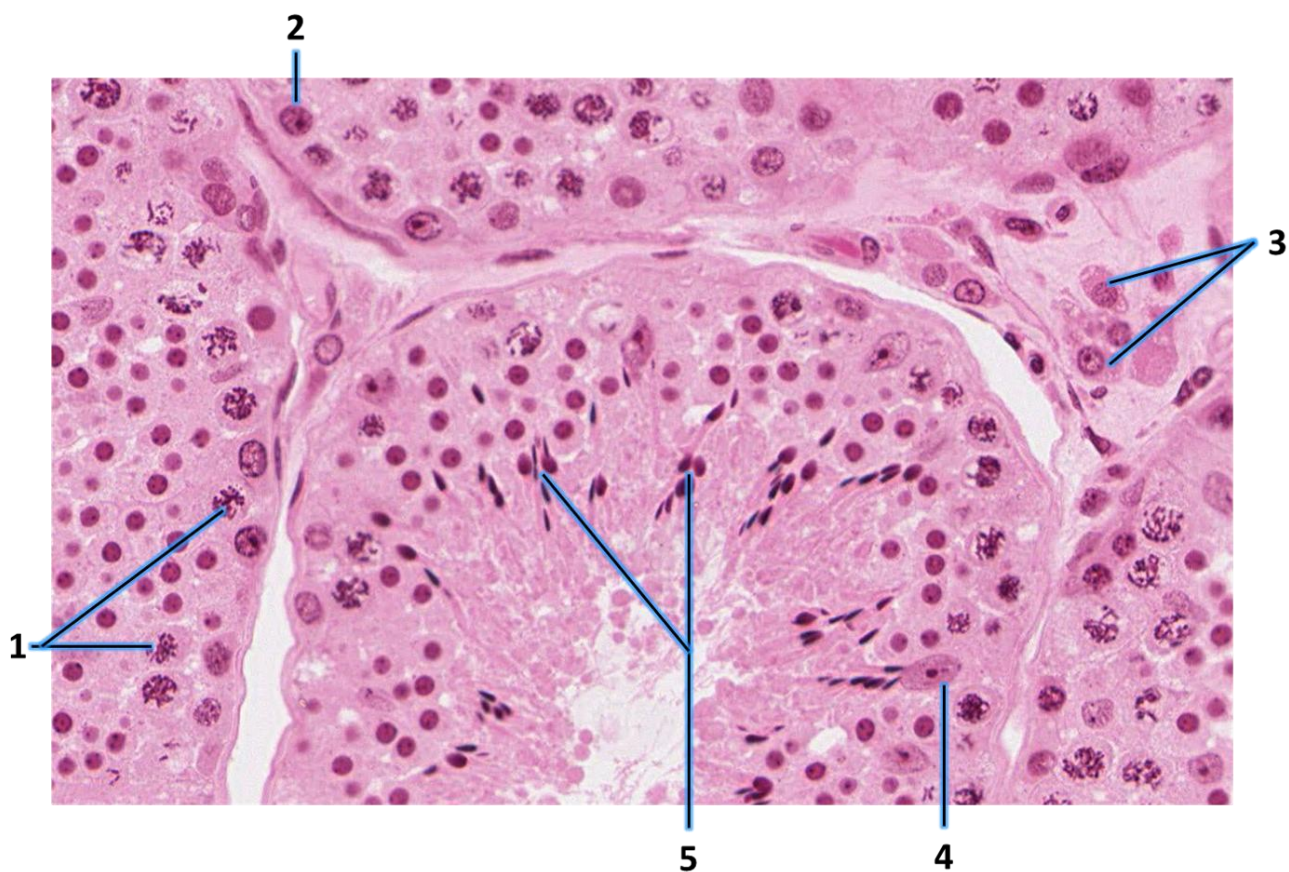
A 25-year-old man comes to the physician with his wife because of a 1-year history of unsuccessful conception. Physical examination shows no abnormalities. Laboratory studies shows reduced levels of testosterone. Which of the following hormones most likely stimulates the production of the affected hormone in this patient?

- A. Estrogen
 - B. Inhibin
 - C. Progesterone
 - D. Follicular stimulating hormone
 - E. Luteinizing hormone
-

Question #: 18

A research group is investigating the effect of a synthetic form of FSH on spermatogenesis and cells of the seminiferous tubules in mice. The drug is administered over a period of two to three months to the subjects following which a marked increase in spermatozoa is observed in the semen. Biopsy samples of testicular tissue is provided in the attached photomicrograph. Which of the following cells is the most likely target of this drug?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5



Question #: 19

A 65-year-old woman comes to the physician because of a 3-month history of bright red blood per vagina and a 6-month history of unintentional weight loss of 13.6-kg (30-lb). She has been menopausal for the past 10 years. An ultrasound scan shows an abnormally thick endometrium. Microscopic examination of a biopsy specimen of the endometrium shows metaplastic changes to the epithelial cells. Which of the following classifications best describes the epithelium in which the metaplastic changes were observed?

- A. Transitional
- B. Simple squamous
- C. Simple columnar
- D. Simple cuboidal

E. Stratified squamous

Question #: 20

A 28-year-old woman who has been diagnosed with leukemia decides to have her oocytes harvested and frozen prior to chemotherapy. Prior to harvesting she is given medications to stimulate follicular growth. Subsequently, oocytes were harvested from pre-ovulatory follicles. The harvested oocytes just prior to ovulation will be in which of the following stages of cell division?

- A. Anaphase of 2nd meiotic division
 - B. Prophase of 1st meiotic division
 - C. Prophase of 2nd meiotic division
 - D. Anaphase of 1st meiotic division
 - E. Metaphase of the 2nd meiotic division
-

Question #: 21

A 30-year-old woman comes to the physician for a scheduled radical hysterectomy with oophorectomy. She has a history of metastatic cancer diagnosed 6-weeks ago. Which of the following hormones is secreted by the cells indicated by the arrow?

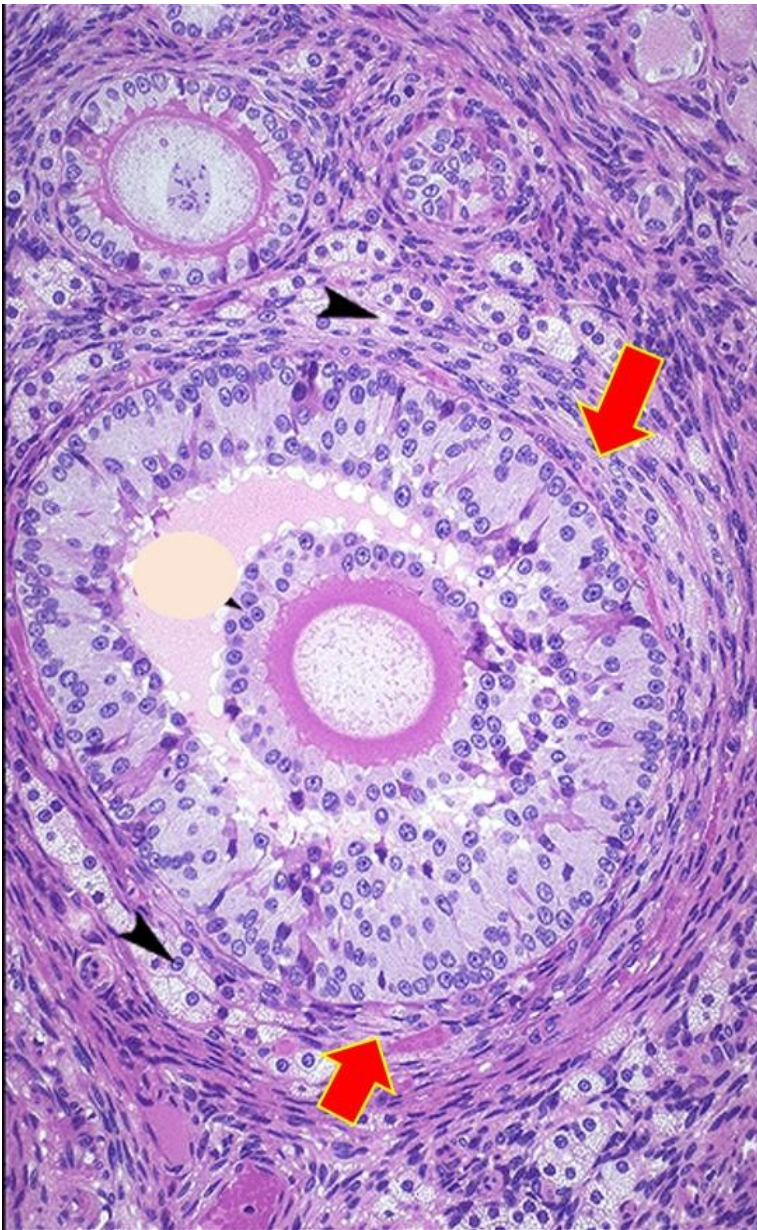
- A. Progesterone
- B. Estrogen
- C. Luteinizing hormone
- D. Androstenedione
- E. Follicle stimulating hormone



Question #: 22

A 45-year-old woman comes to the physician because of a 4-month history of persistent pelvic pain, worse on sexual intercourse and irregular menstrual cycles. Pelvic CT-scan shows a suspicious mass on the right ovary. Microscopic examination of the ovarian mass show abnormal cellular structures. A normal photomicrograph is attached and the cells most affected are highlighted by red arrows. Which of the following best identifies a function of these highlighted cells?

- A. Develop and mature into an egg
- B. Protection of the developing oocyte
- C. Formation of the corpus luteum
- D. Synthesize and secrete androstenedione
- E. Production of estrogen hormone



Question #: 23

A 42-year-old man comes to the physician because of a 4-month history of headache, 6.8-kg (15-lb) weight gain, round face, and excess fat in the back of the neck. His blood pressure is 162/92 mm Hg; other vital signs are within normal limits. Physical examination shows truncal and facial obesity, thin extremities, abdominal striae, and hyperpigmentation of skin. Laboratory studies show elevated blood glucose concentration. MRI of brain is pending. Which of the following serum hormonal

alterations is most likely in this patient?

Option	Cortisol	Adrenocorticotropic hormone (ACTH)	Corticotropin releasing hormone (CRH)
A	Increase	No change	Increase
B	Increase	Decrease	No change
C	Decrease	Increase	Increase
D	Decrease	Increase	Decrease
E	Increase	Increase	Decrease

- A. A
- B. B
- C. C
- D. D
- E. E

Question #: 24

A 32-year-old man is brought to the emergency department 1 hour after he was involved in a motor vehicle collision in which he sustained multiple fractures to the right upper and lower extremities and head. He was traveling at 70 mph when he lost control and hit a concrete wall. He is unconscious and bleeding from his wounds. His temperature is 37.0°C (98.6°F), pulse is 116/min, respirations are 18/min, and blood pressure is 114/72 mm Hg. Physical examination shows abrasions over the face, neck, right shoulder, and thigh. Cranial nerves are intact. He is admitted to the intensive care and monitored. Twelve hours later, his condition deteriorates, and he develops traumatic brain injury. He is under severe stress. Which of the following sets of hormonal alterations is most likely in this patient?

Option	Cortisol	Adrenocorticotropic hormone (ACTH)	Corticotropin releasing hormone (CRH)
A	Increased	Increased	Decreased
B	Increased	Increased	Increased
C	Decreased	Increased	Decreased
D	Decreased	Decreased	Increased
E	Increased	Decreased	Decreased
F	Decreased	Decreased	Decreased

- A. A
- B. B
- C. C
- D. D
- E. E
- F. F

Question #: 25

A 21-year-old woman comes to the physician because of a 3-month history of weight loss despite an increased appetite, sweaty palms, and prominent eyes. Her blood pressure is 140/76 mm Hg, respirations are 18/min, and pulse is 106/min. Physical examination shows a diffuse goiter. A thyroid scan shows increased radio-iodine uptake. She is diagnosed with Graves' disease. Which of the following is most likely expected in this patient?

- A. Decreased coupling of monoiodotyrosine (MIT) and diiodotyrosine (DIT)
 - B. Decreased thyroid hormone binding globulins
 - C. Increased thyroid stimulating hormone (TSH) and thyrotropin releasing hormone (TRH) levels
 - D. Stimulation of the Na/I symporter
 - E. Decreased synthesis of thyroglobulin
-

Question #: 26

A 48-year-old man comes to the physician because of a 6-month history of increasing shoe size and hat size, prominent growth of the nose and thickening of fingers. His blood pressure is 146/91 mm Hg and pulse is 78/min. Physical examination shows a prominent forehead, large nose, protruding mandible, enlarged hands and feet, and coarse skin.

Laboratory studies show elevated fasting blood glucose and plasma insulin-like growth factor-1. Under physiologic conditions, which of the following hormones is most likely released in response to an increased plasma IGF-1?

- A. Growth Hormone
 - B. Growth Hormone Releasing Hormone
 - C. Somatostatin
 - D. Prolactin
 - E. Ghrelin
-

Question #: 27

A 36-year-old man comes to the physician because of a 5-week history of progressive enlargement of his left scrotum. There is no history of infection. His vital signs are within normal limits. Physical examination shows a hard mass in the left scrotum. Testicular cancer is suspected and both testes are surgically removed. Histologic examination of a biopsy specimen of this mass confirms seminoma testis. A decrease in which of the following hormones is most likely in this patient following the surgery?

- A. Gonadotropin releasing hormone (GnRH)
- B. Follicular stimulating hormone (FSH)
- C. Inhibin
- D. Adrenal androgens

Question #: 28

A 22-year-old woman comes to the physician because of a 2-month history of heart palpitations, sweaty palms, rapid weight loss, eyes appearing bigger than usual and feeling anxious and fidgety. Her blood pressure is 136/80 mmHg, pulse is 100/min, and respirations are 14/min. Physical examination shows protruding eyes with a staring appearance. She is suspected of having Graves' disease. The elevated hormone responsible for her symptoms would trigger a response in its target cell by which of the following most likely mechanisms?

- A. IP3 and DAG second messenger system
 - B. cAMP second messenger system
 - C. Cell depolarization
 - D. Nuclear receptor binding
 - E. Tyrosine kinase activation
-

Question #: 29

A 23-year-old man comes to the physician because of a 4-month history of erectile dysfunction. He takes no medications. Physical examination shows no abnormalities. Laboratory studies show normal levels of testosterone, LH and FSH. The physician diagnoses that his erectile dysfunction is related to performance anxiety. He is treated with psychotherapy which normalizes his erectile function. The physiological pathway for penile erection most likely depends on which of the following mechanisms?

- A. Increased inhibin formation in penile tissue
- B. Stimulation of calcium release in penile smooth muscle
- C. Inhibition of the cGMP in penile smooth muscle

- D. Stimulation of the nitric oxide (NO) formation in penile tissue
 - E. Stimulation of cAMP in penile smooth muscle
-

Question #: 30

A 34-year-old woman comes to the physician because of a 6-month history of passing large volume of urine with increased frequency of urination and increased thirst. She was diagnosed with sarcoidosis 4 months ago and is on treatment. Her pulse is 98/min, blood pressure is 112/72 mm Hg, and respirations are 16/min. Physical examination shows increased skin turgor and dry lips and tongue. Laboratory studies show:

Test	Patient value (Reference value)
Serum sodium	154 mEq/L (Normal: 136-146 mEq/L)
Serum calcium	12.8 mg/dL (Normal: 8.5-10.5 mg/dL)
Random blood glucose	90 mg/dL (Normal: 80-120 mg/dL)
Serum osmolality	305 mOsmol/kg H ₂ O (Normal: 275-295 mOsmol/kg H ₂ O)
24-hour urine volume	5800 mL/day (Normal: 800-2000 mL/day)
Urine osmolality	105 mOsmol/kg H ₂ O (Normal: 50-1200 mOsmol/kg H ₂ O)
Urine specific gravity	1.010 (Normal: 1.005-1.030)

She is admitted and a water deprivation test is performed on her. She is restricted from drinking fluids for 2 hours and is administered an injection of desmopressin (synthetic vasopressin). Urine osmolality measured 30 min and 60 min after this injection shows osmolality of 106 mOsmol/kg H₂O and 105 mOsmol/kg H₂O respectively. Which of the following is the most likely diagnosis in this patient?

- A. Psychogenic polydipsia

- B. Syndrome of inappropriate ADH (SIADH)
- C. Nephrogenic diabetes insipidus
- D. Cerebral salt wasting syndrome
- E. Neurogenic diabetes insipidus